# NUMERICAL & STATISTICAL METHODS

## Subject Code : 23AS4406

## UNIT - V

### Small Sample Tests

**Syllabus:** Student t-distribution (test for single mean, two means and paired t-test), testing of equality of variances (F-test), chi-square test for goodness of fit, chi-square test for independence of attributes.

## I. Small Sample Tests : Introduction

When the sample size **$n < 30$**, the normal (Z) test is not reliable because the sampling distribution of the statistic is not normal. We use **exact sampling distributions** : t, F and $\chi^2$ distributions. These need the assumption that the **parent population is normal**.

| Test | Used for | Distribution |
|---|---|---|
| t-test | mean(s) of small samples | Student's t, df = $\nu$ |
| F-test | equality of two variances | Snedecor's F, ($\nu_1$, $\nu_2$) |
| $\chi^2$ test | goodness of fit, independence | Chi-square, df = $\nu$ |

**Degrees of freedom (df)** : number of independent observations available for estimating a quantity = (number of observations) − (number of constraints / estimated parameters).

## 2. Student's t-distribution

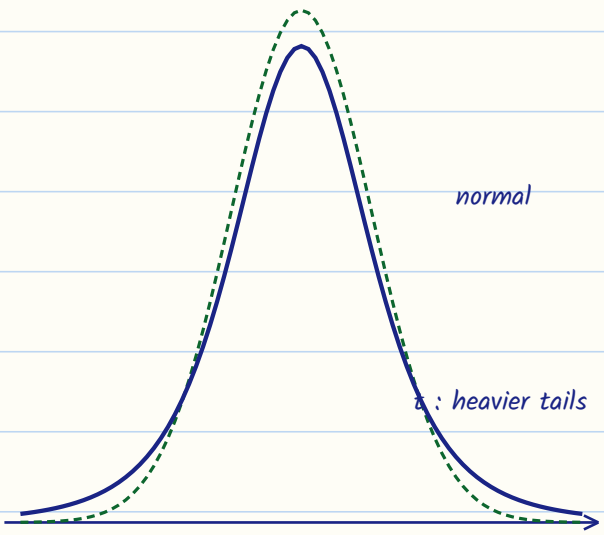
Let $x_1, x_2 \ldots x_n$ be a random sample from $N(\mu, \sigma^2)$. Define

$$t = (\bar{x} - \mu) / (s/\sqrt{n}) \ , \ s^2 = \Sigma (x_i - \bar{x})^2 / (n - 1)$$

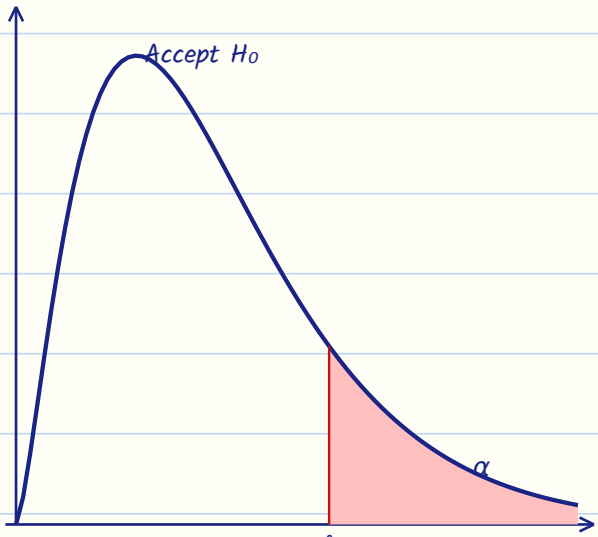
t follows **Student's t-distribution** with $\nu = n - 1$ degrees of freedom. (W. S. Gosset, pen name "Student").

### Properties of the t-distribution

- Bell shaped and **symmetric about t = 0** ; mean = 0 ; variance = $\nu/(\nu - 2)$ for $\nu > 2$.
- Flatter than normal curve with **heavier tails** ; as $\nu \to \infty$ it tends to $N(0, 1)$.
- It depends only on $\nu$ (not on $\mu, \sigma$).
- Total area under the curve = 1 ; critical values are read from t-table for $(\nu, \alpha)$.




t-distribution (n small)




$\chi^2$ distribution (skewed)

## Selected t-table values (two-tailed)

| df | 5 | 7 | 9 | 11 | 15 | 20 | ∞ |
|---|---|---|---|---|---|---|---|
| α = 0.05 | 2.571 | 2.365 | 2.262 | 2.201 | 2.131 | 2.086 | 1.960 |
| α = 0.01 | 4.032 | 3.499 | 3.250 | 3.106 | 2.947 | 2.845 | 2.576 |

## Applications of t

- Test for a single mean ; test for difference of two means (independent) ; paired t-test ; confidence limits for mean ; test for correlation coefficient.

# 3. t-test for Single Mean

$H_0 : \mu = \mu_0$ . $t = (\bar{x} - \mu_0) / (s/\sqrt{n})$ with $n - 1$ df, where $s = \sqrt{[\Sigma(x - \bar{x})^2/(n - 1)]}$.

## Procedure

- $H_0 : \mu = \mu_0$ ; $H_1 : \mu \neq \mu_0$ (or one-sided). Select $\alpha$.
- Compute $\bar{x}$, $s$ and then $t$.
- Compare $|t|$ with the table value $t_\alpha(n - 1)$. If $|t|$ > table value → reject $H_0$.

## Example 1

Ten individuals have the heights 70, 72, 68, 74, 66, 71, 69, 73, 67, 70 (inches). Is it reasonable to believe that the mean height of the population is 68 inches ? (5 %)

$\bar{x} = 700/10 = 70$. Deviations $(x - \bar{x})$ : 0, 2, −2, 4, −4, 1, −1, 3, −3, 0.

$\Sigma(x - \bar{x})^2 = 0 + 4 + 4 + 16 + 16 + 1 + 1 + 9 + 9 + 0 = 60$ ; $s^2 = 60/9 = 6.667$ ; $s = 2.582$

$t = (70 - 68)/(2.582/\sqrt{10}) = 2/0.8165 = \mathbf{2.449}$

$df = 9$, table $t_{0.05} = 2.262$. Since $2.449 > 2.262$, $H_0$ is **rejected** : the mean height is not 68 inches.

### Confidence limits for $\mu$ (small sample)

$\bar{x} \pm t_\alpha \cdot s/\sqrt{n}$. For Example 1 (95 %) : $70 \pm 2.262\ (0.8165) = 70 \pm 1.85 \rightarrow$ **(68.15 , 71.85)**. As 68 is outside, $H_0$ rejected ✓.

## 4. t-test for Difference of Two Means (independent

Two independent small samples from normal populations with **equal variances** $\sigma_1^2 = \sigma_2^2$ . $H_0 : \mu_1 = \mu_2$.

$$t = (\bar{x}_1 - \bar{x}_2) \,/\, [\, s\sqrt{(1/n_1 + 1/n_2)}\,] \text{ with } n_1 + n_2 - 2 \text{ df}$$

$$s^2 = [\, \Sigma(x_1 - \bar{x}_1)^2 + \Sigma(x_2 - \bar{x}_2)^2\,] \,/\, (n_1 + n_2 - 2) \text{ (pooled variance)}$$

### Example 2

Sample 1 : 20, 16, 26, 27, 23, 22 ($n_1 = 6$). Sample 2 : 27, 33, 42, 35, 32, 34, 38 ($n_2 = 7$). Test whether the means differ (5 %).

$\bar{x}_1 = 134/6 = 22.33$ ; $\Sigma(x_1 - \bar{x}_1)^2 = 81.33$

$\bar{x}_2 = 241/7 = 34.43$ ; $\Sigma(x_2 - \bar{x}_2)^2 = 133.71$

$s^2 = (81.33 + 133.71)/11 = 19.55$ , $s = 4.421$

S.E. $= 4.421\sqrt{(1/6 + 1/7)} = 4.421 \times 0.5563 = 2.460$

$t = (22.33 - 34.43)/2.460 = \mathbf{-4.92}$. $df = 11$, $t_{0.05} = 2.201$.

$|t| = 4.92 > 2.201 \rightarrow H_0$ **rejected** : the two means differ

significantly.

> Before using this test one should check that $\sigma_1^2 = \sigma_2^2$ using the **F-test** (next section).

## 5. Paired t-test

Used when the observations are in **pairs on the same units** (before & after treatment, two methods on same subjects). Let $d_i$ = difference of each pair. $H_0 : \mu_d = 0$.

$t = \bar{d} / (s_d/\sqrt{n})$ , $\bar{d} = \Sigma d/n$ , $s_d^2 = \Sigma(d - \bar{d})^2/(n - 1)$ , $df = n - 1$

### Example 3

Marks of 8 students before and after coaching :

| Before | 8 | 10 | 12 | 15 | 11 | 9 | 14 | 13 |
|---|---|---|---|---|---|---|---|---|
| After | 10 | 11 | 14 | 17 | 12 | 10 | 13 | 16 |
| $d$ = After − Before | 2 | 1 | 2 | 2 | 1 | 1 | −1 | 3 |

$\Sigma d = 11$ , $\bar{d} = 1.375$ ; $\Sigma d^2 = 25$ ; $s_d^2 = [25 - 8(1.375)^2]/7 = (25 - 15.125)/7 = 1.4107$ ; $s_d = 1.188$

$t = 1.375/(1.188/\sqrt{8}) = 1.375/0.420 = \mathbf{3.27}$, $df = 7$, $t_{0.05} = 2.365$.

$3.27 > 2.365 \rightarrow H_0$ rejected : coaching **has improved** the marks (significantly).

### Difference between independent and paired t

| Independent t | Paired t |
|---|---|
| Two separate groups | Same group measured twice |
| $df = n_1 + n_2 - 2$ | $df = n - 1$ |
| Uses pooled variance | Uses variance of differences d |

## 6. F-test : Equality of Two Variances

**F-distribution** (Snedecor) : ratio of two independent sample variances from normal populations.

$$F = s_1^2 / s_2^2 \text{ with } (\nu_1, \nu_2) = (n_1 - 1 , n_2 - 1)$$

(take the larger variance in the numerator, so that $F \geq 1$).

$H_0 : \sigma_1^2 = \sigma_2^2$ ; $H_1 : \sigma_1^2 \neq \sigma_2^2$ (or $>$). Reject $H_0$ if $F > F_{_\alpha}(\nu_1, \nu_2)$ from the F-table.

### Properties of F

- Positively skewed, never negative ; shape depends on $(\nu_1, \nu_2)$ ; mean $= \nu_2/(\nu_2 - 2)$ ; $F(\nu_1, \nu_2) = 1/F(\nu_2, \nu_1)$.

### Example 4

Use the two samples of Example 2 to test $\sigma_1^2 = \sigma_2^2$ at 5 %.

$s_1^2 = 81.33/5 = 16.27$ $(n_1 = 6)$ ; $s_2^2 = 133.71/6 = 22.29$ $(n_2 = 7)$.

$F = 22.29/16.27 =$ **1.37** with (6, 5) df. Table $F_{0.05}(6, 5) = 4.95$.

$1.37 < 4.95 \rightarrow H_0$ accepted : the variances are equal, so the pooled t-test in Example 2 was valid ✓.

### Example 5

Two samples : $n_1 = 9$, $s_1^2 = 40$ ; $n_2 = 8$, $s_2^2 = 10$. Is $\sigma_1^2 > \sigma_2^2$ ? (5 %, one-tailed)

$F = 40/10 = 4.0$ with (8, 7) df ; $F_{0.05}(8, 7) = 3.73$. Since $4.0 > 3.73$, **$H_0$ rejected** : $\sigma_1^2$ is greater.

## 7. Chi-Square ($\chi^2$) Test

If O = observed frequency and E = expected frequency in each class then

$$\chi^2 = \Sigma (O - E)^2 / E \text{ with } \nu \text{ degrees of freedom}$$

### Properties of $\chi^2$ distribution

- Values from 0 to $\infty$ ; **positively skewed** ; mean = $\nu$, variance = $2\nu$.
- Sum of two independent $\chi^2$ variables is again $\chi^2$ with df added.
- For large $\nu$, $\chi^2$ tends to normal.

### Conditions for validity

- Total frequency N should be large ($N > 50$) ; each expected frequency $\geq 5$ (otherwise pool neighbouring classes).
- Observations are independent ; expected frequencies sum to total of observed.

| df | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $\chi^2_{0.05}$ | 3.841 | 5.991 | 7.815 | 9.488 | 11.070 | 12.592 |
| $\chi^2_{0.01}$ | 6.635 | 9.210 | 11.345 | 13.277 | 15.086 | 16.812 |

## 7.1 $\chi^2$ test for goodness of fit

$H_0$ : the data follow the specified (theoretical) distribution. df = (number of classes − 1 − number of parameters estimated from data).

### Example 6

A die is thrown 60 times. Observed : 1→8, 2→12, 3→9, 4→11, 5→14, 6→6. Is the die fair ? (5 %)

$H_0$ : die is fair, E = 60/6 = 10 for each face.

| Face | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| O | 8 | 12 | 9 | 11 | 14 | 6 |
| E | 10 | 10 | 10 | 10 | 10 | 10 |
| $(O-E)^2/E$ | 0.4 | 0.4 | 0.1 | 0.1 | 1.6 | 1.6 |

$\chi^2 = 4.2$ , df = 5 , table value 11.07. Since $4.2 < 11.07 \rightarrow H_0$ **accepted** : die is fair.

### Example 7 : genetics ratio 9 : 3 : 3 : 1

Observed 315, 108, 101, 32 (total 556). Expected = 556 × (9, 3, 3, 1)/16 = 312.75, 104.25, 104.25, 34.75.

$\chi^2 = (2.25^2/312.75) + (3.75^2/104.25) + (3.25^2/104.25) + (2.75^2/34.75) = 0.016 + 0.135 + 0.101 + 0.218 =$ **0.470**

df = 3, table 7.815. 0.47 < 7.815 → accepted : data agree with the ratio 9 : 3 : 3 : 1.

## 7.2 $\chi^2$ test for independence of attributes

Data are arranged in a **contingency table** (r rows × c columns). $H_0$ : the two attributes A and B are independent.

Expected frequency $E$ = (row total × column total) / grand total

$df = (r - 1)(c - 1)$. Compute $\chi^2 = \Sigma(O - E)^2/E$.

### Example 8 (2 × 2)

Smoking and lung disease :

| | Disease | No disease | Total |
|---|---|---|---|
| Smokers | 30 | 20 | 50 |
| Non-smokers | 10 | 40 | 50 |
| Total | 40 | 60 | 100 |

$E(30) = 50 \times 40/100 = 20$, $E(20) = 30$, $E(10) = 20$, $E(40) = 30$.

$\chi^2 = 10^2/20 + 10^2/30 + 10^2/20 + 10^2/30 = 5 + 3.33 + 5 + 3.33 =$ **16.67**

$df = (2 - 1)(2 - 1) = 1$, table value 3.841. $16.67 > 3.841 \rightarrow H_0$ rejected : disease and smoking are **dependent (associated)**.

**Short cut for 2 × 2 table** with cells a b / c d : $\chi^2 = N(ad - bc)^2 / [(a + b)(c + d)(a + c)(b + d)]$ Check : $100(1200 - 200)^2/(50 \cdot 50 \cdot 40 \cdot 60) = 16.67$ ✓

### Yates' correction (for 2 × 2 with small cell, < 5)

Replace $|O - E|$ by $|O - E| - ½$ in each term ; $\chi^2 = N(|ad - bc| - N/2)^2 / [(a+b)(c+d)(a+c)(b+d)]$.

## Example 9 (3 × 2)

Education level vs opinion (count of people) :

| | Agree | Disagree | Total |
|---|---|---|---|
| Low | 20 | 30 | 50 |
| Medium | 40 | 30 | 70 |
| High | 40 | 40 | 80 |
| Total | 100 | 100 | 200 |

Expected : Low 25, 25 ; Medium 35, 35 ; High 40, 40.

$\chi^2 = 25/25 + 25/25 + 25/35 + 25/35 + 0 + 0 = 1 + 1 + 0.714 + 0.714 =$ **3.43**

$df = (3 - 1)(2 - 1) = 2$, table 5.991. $3.43 < 5.991 \rightarrow H_0$ accepted : opinion is independent of education level.

# 7A. More Solved Problems

## Example 10 : one-tailed t-test

A sample of 16 bulbs has mean life 53 hours and $s = 6$. Is the mean life greater than 50 hours ? (5 %)

$H_0 : \mu = 50$ ; $H_1 : \mu > 50$ (right-tailed). $t = (53 - 50)/(6/\sqrt{16}) = 3/1.5 =$ **2.0**

$df = 15$ ; one-tailed 5 % critical value = 1.753. Since $2.0 > 1.753 \rightarrow H_0$ **rejected** : mean life is greater than 50.

> For a **one-tailed** test at $\alpha$ use the table value of the two-tailed column $2\alpha$ (i.e. the 10 % two-tailed column for 5 % one-tailed).

## Example 11 : paired t-test (blood pressure)

Six patients : BP before 150, 160, 155, 170, 165, 158 ; after treatment 144, 155, 153, 160, 158, 155. Does the drug reduce BP ? (5 %)

$d$ = before − after = 6, 5, 2, 10, 7, 3 ; $\bar{d} = 33/6 = 5.5$. Deviations : 0.5, −0.5, −3.5, 4.5, 1.5, −2.5 ; $\Sigma = 41.5$

$s_d^2 = 41.5/5 = 8.3$, $s_d = 2.881$. $t = 5.5/(2.881/\sqrt{6}) = 5.5/1.176 =$ **4.68**

$df = 5$, $t_{0.05}$ (two-tailed) = 2.571 ; 4.68 > 2.571 → $H_0$ rejected : the drug **does reduce** blood pressure.

## Example 12 : F-test with sums of squares

Sample 1 : $n_1 = 10$, $\Sigma(x - \bar{x})^2 = 90 \rightarrow s_1^2 = 90/9 = 10$. Sample 2 : $n_2 = 12$, $\Sigma(y - \bar{y})^2 = 121 \rightarrow s_2^2 = 121/11 = 11$.

$F = 11/10 = 1.1$ with (11, 9) df ; $F_{0.05}(11, 9) = 3.10$. 1.1 < 3.10 → equal variances (accepted).

## Example 13 : goodness of fit to a binomial law

4 coins are tossed 160 times. Number of heads 0, 1, 2, 3, 4 occurred 6, 32, 58, 50, 14 times. Is the coin fair ?

For a fair coin the frequencies are $160 \times (1, 4, 6, 4, 1)/16 = 10, 40, 60, 40, 10$.

| Heads | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| O | 6 | 32 | 58 | 50 | 14 |
| E | 10 | 40 | 60 | 40 | 10 |
| $(O-E)^2/E$ | 1.6 | 1.6 | 0.067 | 2.5 | 1.6 |

$\chi^2 = 7.37$ , $df = 4$ , table 9.488. $7.37 < 9.488$ → accepted : the coins are fair (binomial fits).

## Which test to use ? (decision guide)

- Comparing a **mean** with a standard value : $n \geq 30$ → Z ; $n < 30$ → t (single mean).
- Comparing **two means** : n large → Z ; small independent samples → t (pooled) ; same units twice → paired t.
- Comparing **two variances** → F.
- Comparing **observed with expected counts** or checking association between **attributes** → $\chi^2$.

## Z-test vs t-test

| Z-test | t-test |
|---|---|
| $n \geq 30$ (large) | $n < 30$ (small) |
| Uses $\sigma$ (or s) | Uses s with $(n - 1)$ df |
| Normal distribution | Student's t (heavier tails) |
| Critical value fixed 1.96 | Depends on df (2.262 for 9 df) |

## Assumptions of small sample tests

- Population is (approximately) normal ; samples are random and independent (except paired).
- For the two-sample t-test, population variances are equal (check by F).
- $\chi^2$ test is a **non-parametric** (distribution-free) test, no normality needed.

## Relation between t, F and $\chi^2$

- If t has $\nu$ df then $t^2$ follows $F(1, \nu)$.
- F is the ratio of two independent $\chi^2$ variables each divided by its df : $F = (\chi_1^2/\nu_1)/(\chi_2^2/\nu_2)$.
- $(n-1)s^2/\sigma^2$ follows $\chi^2$ with $(n-1)$ df.

## 8. Quick Revision

| Test | Statistic | df |
|---|---|---|
| t (single mean) | $(\bar{x} - \mu)/(s/\sqrt{n})$ | $n - 1$ |
| t (two means) | $(\bar{x}_1 - \bar{x}_2)/(s\sqrt{(1/n_1 + 1/n_2)})$ | $n_1 + n_2 - 2$ |
| paired t | $\bar{d}/(s_d/\sqrt{n})$ | $n - 1$ |
| F | $s_1^2/s_2^2$ (larger on top) | $(n_1 - 1, n_2 - 1)$ |
| $\chi^2$ | $\Sigma (O - E)^2/E$ | $k - 1$ (fit), $(r-1)(c-1)$ (independence) |

## Exam questions

- State properties of t-distribution. Explain paired t-test with an example.
- Two samples of sizes 9 and 8 have sums of squared deviations 160 and 91 : test equality of variances using F-test ($F = 20.0/13.0 = 1.54$, accept).
- A die is thrown 120 times and the results are ... test goodness of fit using $\chi^2$.
- Test independence of attributes for the 2 × 2 table given (use $\chi^2$).